Windows Firewall GPO to Intune Migration

Implementation Solution Guide

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Executive Summary — Windows Firewall GPO to Intune Migration

White-label · Migrations

Moving Windows Defender Firewall from Group Policy to Microsoft Intune is a known minefield. Do it before the foundation is ready and you get policy conflicts or merge/loop conditions that leave devices in an unpredictable state — the firewall is on, but nobody is sure which rules are actually enforced. Because the firewall governs connectivity, a botched cutover does not just misconfigure a setting; it can cut a fleet off from line-of-business apps, remote management, or each other.

This is a complete, deploy-it-today package for the controlled migration of Windows Defender Firewall from GPO to Intune endpoint-security firewall policy, written for an administrator with no prior migration experience. It does not re-teach net-new firewall design — it is the disciplined GPO-to-Intune move: inventory the GPO firewall rules, translate them into Intune firewall policy and firewall rule profiles, manage GPO/Intune coexistence so the two never fight, cut over in rings with parity validation, and only then retire the GPO firewall settings.

What you get

• A plain-language Solution Overview of the migration model, GPO-to-Intune rule mapping, and the conflict/merge/loop dynamics this migration is built to avoid.

• A Requirements Guide with verify / implement-if-missing for every prerequisite.

• An Implementation Guide of precise, ordered Intune steps: inventory, translate, configure coexistence, assign in rings, validate parity, retire GPO, expand.

• Five PowerShell scripts for firewall-rule discovery, post-migration rule-parity validation, a coverage and conflict report, safe conflict remediation, and a migration-ring group builder.

• Testing and Troubleshooting guides with explicit pass/fail gates and conflict/loop diagnosis.

What this delivers

A ringed firewall migration — `Migration Ring – Pilot`, `Migration Ring – Wave 1`, and `Migration Ring – Broad` — in which GPO firewall rules are translated into a named Intune `Firewall Policy – Migrated Rules – Pilot` and matching `Firewall Rules` profiles, GPO and Intune coexist without conflict during transition, rule parity is validated on every ring before the next, and the GPO firewall settings are retired only after parity holds.

The standard deployment model

Assess → Configure → Pilot → Validate → Expand — with a parity gate and a rollback path at every ring, because the firewall affects connectivity.

Outcome

Windows Defender Firewall moved from GPO to Intune — rules translated, conflicts and loops avoided through managed coexistence, cut over in rings with parity validated, and the legacy GPO firewall settings cleanly retired — proven on a pilot before you scale.

1. Solution Overview — Windows Firewall GPO to Intune Migration

What this migration is

This solution moves Windows Defender Firewall management from Active Directory Group Policy (GPO) to Microsoft Intune endpoint-security firewall policy. It is not net-new firewall design — it assumes you already have working firewall rules in GPO and need to relocate that enforcement to Intune without breaking connectivity. The deliverable is a controlled, ringed cutover: inventory what GPO enforces, translate it into Intune firewall objects, run both side by side without conflict during transition, validate rule parity, then retire the GPO firewall settings.

Why move firewall management to Intune

- Cloud-managed, internet-facing devices do not reliably receive GPO; a domain-joined-only firewall

policy leaves remote and Entra-joined devices unmanaged or inconsistently managed.

- Intune gives a single endpoint-security surface (`Endpoint security` → `Firewall`) with assignment

by Entra group, per-device reporting, and no dependency on line-of-sight to a domain controller.

- Co-management and modern provisioning (Autopilot, Entra join) expect endpoint security to live in

Intune, not GPO.

Why this migration is risky (the *why* behind the discipline)

Group Policy and Intune can both write Windows Defender Firewall settings to the same device. When both are active and unmanaged, three failure modes appear:

- Conflict — GPO and Intune set the same firewall setting to different values; the device's effective

state becomes ambiguous and can flip between policy refreshes.

- Merge condition — local, GPO, and Intune (MDM) rules all merge into the effective rule set. If you

do not control rule merging, you get unexpected union behavior: rules you thought you removed are still enforced from another source.

- Merge loop — GPO re-asserts a firewall setting on its refresh cycle while Intune re-asserts a

conflicting value on its check-in; the device oscillates and never settles, and reporting shows intermittent compliance. This is the pain point this solution is built to avoid.

Because the firewall governs connectivity, an unmanaged conflict does not fail safe — it can block inbound management, RDP/Remote Help, or LOB traffic. That is why the migration is staged (inventory → translate → coexist → ring → validate → retire) rather than a flip of a switch.

How GPO firewall rules map to Intune

GPO firewall concept	Intune endpoint-security equivalent
Firewall profile state (Domain/Private/Public on, default inbound/outbound action, stealth, logging)	A `Firewall Policy` (profile `Windows Firewall`) — one policy enabling the three profiles with the same defaults
Individual inbound/outbound allow/block rules	`Firewall Rules` profile entries (one rule each: name, direction, action, protocol/ports, program, address scope)
GPO scoping by OU / security group	Intune assignment to an Entra security group (the migration ring)
GPO refresh / gpupdate	Intune device check-in / sync

A single GPO firewall section typically becomes one `Firewall Policy` (the profile defaults) plus one or more `Firewall Rules` profiles (the explicit rules), assigned to the same ring group.

The coexistence model (how conflicts/loops are avoided)

During transition, every in-scope device is targeted by GPO and Intune at once. The migration controls this so they never fight:

1. Translate first, assign nothing — build the Intune objects but do not assign them until the device is ready to leave GPO.
1. Move a device out of GPO firewall scope at the same moment it enters the Intune ring — one authority per device, never two enforcing the same firewall settings.
1. Control rule merging explicitly in the Intune policy so local/GPO leftovers cannot silently re-merge.
1. Validate parity (the Intune effective rule set equals the old GPO rule set) before advancing.

The named artifacts this solution creates

All objects follow `<Purpose> – <Scope> – <Stage>` and are reused verbatim throughout the guides:

- Groups: `FW Migration – Pilot Devices – Stage Pilot`, `FW Migration – Wave 1 Devices – Stage Wave1`, `FW Migration – Fleet Devices – Stage Broad`.
- Firewall policy (profile defaults): `Firewall Policy – Migrated Rules – Pilot`, `Firewall Policy – Migrated Rules – Broad`.
- Firewall rules profiles: `Firewall Rules – Migrated Inbound – Pilot`, `Firewall Rules – Migrated Outbound – Pilot` (and the Broad equivalents).

• Migration rings: `Migration Ring – Pilot`, `Migration Ring – Wave 1`, `Migration Ring – Broad`.

The standard deployment model

> *Assess (inventory GPO firewall rules) → Configure (translate to Intune policy + rules, set coexistence) → Pilot (`Migration Ring – Pilot`) → Validate (rule parity + connectivity) → Expand (waves, then retire GPO). repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Windows Firewall GPO to Intune Migration

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-migrations.md` (target tenant, Intune licensing/admin, source-platform access, inventory, enrollment, pilot/wave structure, change control + coexistence + rollback) — confirm those first.

2.1 Intune endpoint-security admin access

Description: Rights to create and assign `Endpoint security` → `Firewall` policies in Intune. **Why:** The migration target is an Intune firewall policy and firewall rule profiles; you cannot build them without these rights. **Verify:** `https://intune.microsoft.com` → `Endpoint security` → `Firewall` loads and `Create Policy` is available; you hold **Intune Administrator** or **Endpoint Security Manager**. **Implement:** Assign the Intune/Endpoint Security admin role (least privilege) to the migrator.

2.2 Source GPO firewall rule access and export

Description: Read access to the Group Policy Object(s) that configure Windows Defender Firewall profiles and rules for the in-scope devices. **Why:** The migration begins by inventorying exactly what GPO enforces so it can be translated. **Verify:** In Group Policy Management you can open the GPO and see `Computer Configuration` → `Windows Settings` → `Security Settings` → `Windows Defender Firewall with Advanced Security`; or run `FWMIG-Discovery.ps1` on a domain-joined in-scope device (it exports GPO + applied firewall rules, read-only). **Implement:** Obtain delegated read on the firewall GPO(s) from the AD/GPO owner.

2.3 Targeted devices enrolled in Intune

Description: In-scope devices enrolled in Intune (co-managed or Intune-managed). **Why:** A `Firewall Policy` and `Firewall Rules` profiles apply only to enrolled devices. **Verify:** Intune → `Devices` → `All devices` shows the in-scope devices. **Implement:** Enable automatic enrollment / co-management; enroll the devices.

2.4 Supported Windows OS and MDM firewall CSP

Description: Windows 10/11 (Pro/Enterprise/Education) that supports the Firewall configuration service provider Intune uses. **Why:** Intune firewall policy and rules require a supported build to apply. **Verify:** `winver` on a sample device; run `FWMIG-Discovery.ps1` (reports OS build). **Implement:** Patch/upgrade

devices to a supported build before migrating them.

2.5 A documented GPO firewall rule inventory (source of truth)

Description: An accurate export of the GPO firewall profile defaults and every explicit rule for the migration scope. **Why:** Parity validation and GPO retirement depend on knowing exactly what is in scope. **Verify:** You can produce a rule export (CSV/JSON) from `FWMIG-Discovery.ps1` and `FWMIG-Report.ps1`. **Implement:** Run `FWMIG-Discovery.ps1` across a representative device set to build the inventory before scoping rings.

2.6 Coexistence control — one firewall authority per device

Description: A defined approach so that, for any device, GPO **or** Intune enforces the firewall — never both at once on the same settings. **Why:** Two authorities on the same firewall settings cause conflicts and merge loops (Overview). **Verify:** For each ring you can both remove devices from GPO firewall scope and add them to the Intune ring group at the same change step; run `FWMIG-Report.ps1` to flag devices currently double-managed. **Implement:** Use a security-group-filtered GPO (deny-apply or remove from OU scope) keyed to the same membership the ring group uses, so moving a device flips authority atomically.

2.7 Migration ring groups (pilot + waves)

Description: Entra security groups for `Migration Ring – Pilot`, `Migration Ring – Wave 1`, and `Migration Ring – Broad`. **Why:** The firewall is cut over in rings with a parity gate between each; connectivity risk demands it. **Verify:** Entra → `Groups`; the three groups exist (or run `FWMIG-Set-MigrationRingTargeting.ps1` to create/populate them). **Implement:** Create the groups (`FWMIG-Set-MigrationRingTargeting.ps1` or Entra → `Groups` → `New group` → Security).

2.8 Change control + rollback (connectivity-aware)

Description: A change window and a written rollback (re-scope the device to GPO firewall / unassign the Intune policy) for each ring, because the firewall affects connectivity. **Why:** A bad rule translation can block management or LOB traffic; cutover must be reversible. **Verify:** A rollback procedure is written and approved before any ring cutover. **Implement:** Define per-ring rollback, comms, and the order coexistence → cutover → retire (see §G7 of the global requirements).

Checklist

- Intune endpoint-security admin access
- Source GPO firewall rules readable / exportable — `FWMIG-Discovery.ps1`
- In-scope devices enrolled in Intune
- Supported Windows build (firewall CSP) — `FWMIG-Discovery.ps1`
- GPO firewall rule inventory captured — `FWMIG-Report.ps1`
- Coexistence control (one authority per device) defined
- `Migration Ring – Pilot` / `Wave 1` / `Broad` groups created

• Change control + connectivity-aware rollback approved

3. Implementation Guide — Windows Firewall GPO to Intune Migration

Follow these steps **in order**. Intune work is in the **Microsoft Intune admin center** (`https://intune.microsoft.com`); inventory work uses the included scripts on a domain-joined, in-scope device. Reuse the named artifacts from the Overview verbatim.

> ***Model:*** *Inventory GPO firewall rules → translate to an Intune* `Firewall Policy` *plus* `Firewall Rules` *profiles → set coexistence so GPO and Intune never enforce the same device at once → assign in rings with a parity gate → retire GPO firewall settings → expand.* ***Assess → Configure → Pilot → Validate → Expand.***

Step 1 — Assess: inventory the GPO firewall rules

1. On a domain-joined in-scope device, run `FWMIG-Discovery.ps1`. It exports the applied

Windows Defender Firewall profile defaults and every rule (read-only) using the NetSecurity module (`Get-NetFirewallProfile`, `Get-NetFirewallRule`) and the GPO-sourced rule store.

1. Run `FWMIG-Report.ps1` to produce the coverage + conflict report — the source-of-truth rule

inventory you will validate parity against, plus any device already double-managed.

1. Confirm the inventory matches the GPO in Group Policy Management

(`Computer Configuration` → `Windows Settings` → `Security Settings` → `Windows Defender Firewall with Advanced Security`).

Step 2 — Create the migration ring groups

1. Run `FWMIG-Set-MigrationRingTargeting.ps1` to create/populate

`Migration Ring – Pilot`, `Migration Ring – Wave 1`, and `Migration Ring – Broad`; or in Entra → `Groups` → `New group` → Security create them manually.

1. Put a few representative devices in `Migration Ring – Pilot` (varied locations/roles so connectivity

edge cases surface early).

Step 3 — Configure: translate the profile defaults to a Firewall Policy

1. Intune → `Endpoint security` → `Firewall` → `Create Policy`.
2. **Platform:** Windows. **Profile:** `Windows Firewall`. **Create**.
3. **Name:** `Firewall Policy – Migrated Rules – Pilot`. **Next**.
4. On `Configuration settings`, for **Domain**, **Private**, and **Public** profiles, set the values

read from the GPO inventory (Step 1):

- `Firewall enabled` = `Allowed`.
- `Default inbound action` = match the GPO (typically `Block`).
- `Default outbound action` = match the GPO (typically `Allow`).
- `Stealth mode`, `Inbound notifications`, `Unicast responses to multicast broadcast`, logging path/size and `Log dropped packets` = match the GPO values.

1. Set the rule-merge controls explicitly so leftovers cannot silently re-merge (key to avoiding merge conditions): set the policy/local rule-merge settings to the values your coexistence plan requires — during transition do not allow uncontrolled merging of local or GPO rules.
1. **Do not assign yet.** Click **Next** through assignments leaving them empty, then **Create**.

Step 4 — Configure: translate the explicit rules to Firewall Rules profiles

1. `Endpoint security` → `Firewall` → `Create Policy` → Windows → `Windows Firewall Rules`. **Create**.
2. **Name:** `Firewall Rules – Migrated Inbound – Pilot`.
3. For each inbound rule in the inventory, add one rule entry: `Name`, `Direction` = In, `Action` (Allow/Block), `Protocol`/`Ports`, `Program`/service, and `Local`/`Remote address` scope — mirroring the GPO rule exactly. `FWMIG-Report.ps1` lists each rule's fields to transcribe.
1. **Do not assign yet. Create**.
2. Repeat for outbound rules as `Firewall Rules – Migrated Outbound – Pilot` (omit if the GPO has no explicit outbound rules).

Step 5 — Configure coexistence (avoid conflicts and merge loops)

1. Confirm the coexistence control from Requirement 2.6: a device leaves GPO firewall scope at the same moment it joins `Migration Ring – Pilot` (group-filtered GPO deny-apply, or OU move keyed to the same membership). One authority per device.
1. Run `FWMIG-Report.ps1` and confirm no pilot device is double-managed before assigning.

Step 6 — Pilot: assign to the first ring

1. Open `Firewall Policy – Migrated Rules – Pilot` → `Assignments` → `Add groups` → `Migration Ring – Pilot`. **Save**.
1. Assign `Firewall Rules – Migrated Inbound – Pilot` (and the outbound profile) to `Migration Ring – Pilot`.
1. As each device joins the ring, the coexistence control removes it from GPO firewall scope. On a pilot device, sync policy (`Settings` → `Accounts` → `Access work or school` → `Info` → Sync).

Step 7 — Validate: rule parity and connectivity (the gate)

1. Run `FWMIG-Validation.ps1` on pilot devices → confirms the **effective** firewall rule set now

matches the captured GPO inventory (profile defaults + each rule), and that GPO is no longer enforcing firewall settings on that device.

1. Run `FWMIG-Report.ps1` → coverage report shows pilot devices fully translated with no conflicts.
2. Verify connectivity: required inbound services (RDP/Remote Help, LOB listeners) and outbound paths

still work. Complete the **Testing Guide**.

1. **Parity gate:** do not advance until parity passes and connectivity is confirmed on the pilot.

If it fails, use the rollback (re-scope the device to GPO firewall / unassign the Intune policy).

Step 8 — Retire the GPO firewall settings (for migrated devices only)

1. Once a device has passed parity in Intune and is out of GPO firewall scope, it no longer needs the

GPO firewall section — but do not delete the GPO until **all** rings are migrated.

1. Re-run `FWMIG-Report.ps1` to confirm no remaining device depends on the GPO firewall settings,

then disable/remove the firewall section from the GPO (or unlink it) as the final retirement step.

Step 9 — Expand: waves, then broad

1. Move to `Migration Ring – Wave 1`: clone the pilot objects as

`Firewall Policy – Migrated Rules – Broad` / `Firewall Rules – Migrated * – Broad` (or reuse the pilot policy by adding the wave group to its assignments), assign, validate parity, retire GPO scope.

1. Repeat for `Migration Ring – Broad` (the rest of the fleet), passing the parity gate each ring.
2. Only after the final ring passes parity, retire the GPO firewall settings (Step 8) fleet-wide.

Order of operations

`Inventory GPO rules → ring groups → translate to Firewall Policy → translate to Firewall Rules → set coexistence → assign Pilot → validate parity → retire GPO scope → expand (waves → Broad) → retire GPO`

5. Testing Guide — Windows Firewall GPO to Intune Migration

Run after each ring is assigned (Implementation Step 7) and again before retiring any GPO firewall settings. The firewall affects connectivity, so every ring must pass before the next.

Test 1 — Intune policy applied

Do: Intune → `Endpoint security` → `Firewall` → `Firewall Policy – Migrated Rules – Pilot` → `Device status`. **Expected:** Ring devices show **Succeeded** (also for the `Firewall Rules` profiles).

Test 2 — Profile defaults match the GPO

Do: On a ring device run `FWMIG-Validation.ps1`. **Expected:** Domain/Private/Public enabled state and default inbound/outbound actions equal the captured GPO inventory values.

Test 3 — Rule parity (effective rule set equals the GPO inventory)

Do: Run `FWMIG-Validation.ps1` (compares the effective `Get-NetFirewallRule` set against the captured GPO rule inventory). **Expected:** Every migrated rule is present with matching direction, action, protocol/ports, program, and address scope; no expected rule missing, no stray rule.

Test 4 — Single authority (no double management)

Do: Run `FWMIG-Report.ps1` for the ring; on a device check the rule source via `FWMIG-Validation.ps1`. **Expected:** Firewall settings come from Intune (MDM) only; the device is out of GPO firewall scope — no conflict, no merge loop, stable compliance across two check-ins.

Test 5 — Connectivity preserved

Do: Exercise required inbound services (e.g. RDP/Remote Help, a LOB listener) and outbound paths from a ring device. **Expected:** All traffic that worked under GPO still works under Intune; nothing newly blocked.

Test 6 — No merge-loop oscillation

Do: Re-run `FWMIG-Validation.ps1` after two device check-ins / a `gpupdate /force` on a migrated device. **Expected:** Effective firewall state is identical both times — no flipping values, no reintroduced GPO rules.

Test 7 — Coverage report

Do: Run `FWMIG-Report.ps1` for the ring. **Expected:** A clear per-device coverage and conflict report — devices translated, parity status, and zero conflicts before the gate.

Pass/fail gate

Gate	Requirement
Ring parity pass	Tests 1–4 and 6 pass; effective Intune rule set equals the GPO inventory with one authority
Connectivity pass	Test 5 passes — no required service newly blocked
Retire-GPO pass	Test 7 shows no device depends on the GPO firewall settings (all rings migrated)

6. Troubleshooting Guide — Windows Firewall GPO to Intune Migration

Use `FWMIG-Discovery.ps1` and `FWMIG-Validation.ps1` for per-device state, `FWMIG-Report.ps1` for fleet coverage/conflicts, and `FWMIG-Remediation.ps1` (default `-WhatIf`) for safe fixes.

A. Conflict — GPO and Intune disagree on a setting

Symptom	Cause	Remediation
Effective default inbound/outbound action differs from the Intune policy	The device is still in GPO firewall scope while the Intune policy is assigned (two authorities)	Run `FWMIG-Report.ps1` (flags double-managed); remove the device from GPO firewall scope so only Intune enforces; re-sync
Intune policy shows Succeeded but the device behaves differently	GPO refresh re-asserts the old value after Intune applies	Confirm the coexistence control moved the device out of GPO scope at ring entry; `FWMIG-Remediation.ps1` to confirm/fix authority

B. Merge condition — unexpected rules still enforced

Symptom	Cause	Remediation
A rule you "removed" is still active	Local/GPO rules are merging into the effective set	Set the rule-merge controls in `Firewall Policy – Migrated Rules – *` to disallow uncontrolled local/GPO merging; re-validate with `FWMIG-Validation.ps1`
Effective rule set is a superset of the Intune rules	Multiple rule sources merged	Identify the extra rules' source with `FWMIG-Validation.ps1` (reports rule origin); remove the stray source

C. Merge loop — oscillating / intermittent compliance

Symptom	Cause	Remediation
Compliance flips between check-ins; effective firewall state changes each cycle	GPO re-asserts on its refresh while Intune re-asserts on check-in — the two are fighting	This is the classic loop: ensure exactly one authority per device. Use `FWMIG-Report.ps1` to find double-managed devices, remove them from GPO firewall scope, then `FWMIG-Validation.ps1` over two check-ins / a `gpupdate /force` to confirm the state is now stable

Symptom	Cause	Remediation
`FWMIG-Validation.ps1` returns different results on consecutive runs	Same as above	Do not advance the ring until the state is stable; fix coexistence first

D. Parity failure — Intune rule set != GPO inventory

Symptom	Cause	Remediation
A migrated rule is missing	Rule not transcribed into the `Firewall Rules` profile	Add the missing rule from the `FWMIG-Report.ps1` inventory; re-validate
A rule's behavior differs	Field mistranslated (protocol/ports/program/address scope)	Compare the `FWMIG-Validation.ps1` diff to the inventory; correct the rule field

E. Connectivity broken after cutover

- A required inbound service (RDP/Remote Help, LOB listener) or outbound path stopped working →

the corresponding allow rule was not translated or was mistranslated. Find it in the `FWMIG-Report.ps1` inventory, add/fix the rule in the `Firewall Rules` profile, re-validate. If it is blocking management, use the ring rollback (re-scope to GPO firewall / unassign Intune) and fix before re-attempting.

F. Policy not applying at all

- Device not enrolled / not in the ring group → confirm enrollment (Intune → `Devices`) and

membership of `Migration Ring – Pilot` (or current ring); sync.

- Unsupported build → `FWMIG-Discovery.ps1` reports the OS build; patch/upgrade.

G. Diagnostics quick reference

```
Get-NetFirewallProfile | Select Name, Enabled, DefaultInboundAction, DefaultOutboundAction
Get-NetFirewallRule -PolicyStore ActiveStore | Select DisplayName, Direction, Action, Enabled
Get-NetFirewallRule -PolicyStore ActiveStore | Get-NetFirewallRule  # inspect effective (merged) store
gpresult /h gpo-firewall.html   # confirm whether the firewall GPO still applies to the device
gpupdate /force                 # force a GPO refresh to test for oscillation
```

Effective rules live in the `ActiveStore`; GPO-sourced rules in the RSOP/GPO store — comparing the two shows whether GPO is still contributing.

H. When to escalate

If exactly one authority enforces the firewall (no double management), rule parity passes, and the device is synced, but connectivity still fails or compliance still oscillates, capture `FWMIG-Validation.ps1` output plus `gpresult /h` and open a Microsoft support case.

7. Best Practices — Windows Firewall GPO to Intune Migration

• **Inventory before you translate.** Capture the GPO firewall profile defaults and every rule with `FWMIG-Discovery.ps1` / `FWMIG-Report.ps1` first; that export is your parity source of truth and your GPO-retirement checklist.

• **One authority per device, always.** The single most important practice: never let GPO and Intune enforce the same firewall settings on the same device at once. Move a device out of GPO firewall scope at the same step it enters the Intune ring.

• **Control rule merging explicitly.** Set the merge controls in the Intune `Firewall Policy` so local and GPO leftovers cannot silently re-merge — uncontrolled merging is how merge conditions and loops start.

• **Ring it; never big-bang the firewall.** `Migration Ring – Pilot` → `Wave 1` → `Broad`, with a parity gate between each. The firewall affects connectivity, so a small ring catches a bad translation before the fleet feels it.

• **Gate on parity, not on "it deployed."** Advance only when the effective Intune rule set equals the captured GPO inventory and connectivity is confirmed — not just when the policy shows Succeeded.

• **Translate faithfully, then improve later.** Migrate rules as-is for parity; defer firewall redesign/cleanup to a separate change after the migration is complete, so a connectivity break is unambiguously a translation error.

• **Keep a connectivity-aware rollback ready.** For each ring, pre-write how to re-scope the device to GPO firewall / unassign the Intune policy, because a wrong rule can block management or LOB traffic.

• **Watch for oscillation.** Re-validate after two check-ins / a `gpupdate /force`; a flipping effective state means two authorities are still fighting — fix coexistence before expanding.

• **Retire GPO last.** Disable/unlink the GPO firewall section only after every ring has passed parity and `FWMIG-Report.ps1` shows no device depends on it.

• **Name everything consistently.** Use `<Purpose> – <Scope> – <Stage>` for every policy, rules profile, ring, and group so the migration state is self-documenting and auditable.

8. Deliverables Summary — Windows Firewall GPO to Intune Migration

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box

Document	Purpose
1. Solution Overview	The migration model, GPO-to-Intune rule mapping, conflict/merge/loop dynamics, coexistence, and the named artifacts
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Migrations)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps: inventory, translate, coexist, ring, validate parity, retire GPO, expand
5. Testing Guide	Parity/connectivity validation with pass/fail gates
6. Troubleshooting Guide	Field issues, including conflict/merge/loop diagnosis
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`FWMIG-Discovery.ps1`	Read-only inventory of GPO + applied Windows Defender Firewall profile defaults and rules (NetSecurity module)
`FWMIG-Validation.ps1`	Read-only post-migration rule-parity check: effective rule set vs the captured GPO inventory, plus authority/source
`FWMIG-Report.ps1`	Read-only coverage + conflict report across devices (local NetSecurity + Intune firewall policy via Microsoft Graph)
`FWMIG-Remediation.ps1`	Safe (`-WhatIf` by default) resolution of conflicts / double-management — proposes fixes, applies only when confirmed
`FWMIG-Set-MigrationRingTargeting.ps1`	Creates/populates the `Migration Ring – Pilot` / `Wave 1` / `Broad` Entra groups (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**

with a parity gate and rollback at every ring.

Outcome

Windows Defender Firewall migrated from GPO to Intune — rules translated and validated for parity, conflicts and merge loops avoided through managed coexistence, cut over in rings, and the legacy GPO firewall settings cleanly retired.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

FWMIG-Discovery.ps1

Purpose: Captures the source-of-truth firewall inventory for a GPO-to-Intune migration: the Domain/Private/Public profile defaults and every applied firewall rule on an in-scope, domain-joined device. This export is what later parity validation is compared against. This script is READ-ONLY - it makes no changes.

How to run: Run on a domain-joined, in-scope device in an elevated PowerShell session: .\FWMIG-Discovery.ps1 .\FWMIG-Discovery.ps1 -OutputPath C:\Temp\fwmig Uses the built-in NetSecurity module (no install). Requires local admin to read the effective firewall policy stores.

FWMIG-Remediation.ps1

Purpose: Resolves the classic GPO-to-Intune migration conflict on a device that is double-managed: when a device has been assigned the Intune (MDM) firewall policy but GPO is STILL applying firewall rules, the two authorities fight and can merge-loop. This script forces a Group Policy refresh and an Intune sync so the device drops out of GPO scope (per your coexistence control) and settles on the single Intune authority, then re-checks the state.

How to run: Preview (default, safe): .\FWMIG-Remediation.ps1 Apply the refresh/sync: .\FWMIG-Remediation.ps1 -Apply

FWMIG-Report.ps1

Purpose: Produces the migration coverage and conflict report: lists the Intune endpoint-security firewall policies and firewall rule profiles (the migration targets) and their assignments via Microsoft Graph, and - on the local device - flags double-management (both GPO and Intune/MDM enforcing the firewall), which is the precursor to conflicts and merge loops. This script is READ-ONLY - it queries only; it makes no changes.

How to run: Connect to Microsoft Graph first (delegated, least privilege): Connect-MgGraph -Scopes 'DeviceManagementConfiguration.Read.All','Group.Read.All' Then: .\FWMIG-Report.ps1 .\FWMIG-Report.ps1 -OutputPath C:\Temp\FWMIG Requires the Microsoft.Graph PowerShell SDK. The local double-management check uses the built-in NetSecurity module and should run on a representative in-scope device.

FWMIG-Set-MigrationRingTargeting.ps1

Purpose: Builds the targeting backbone for the ringed firewall cutover: the Entra security groups 'Migration Ring - Pilot' 'Migration Ring - Wave 1' 'Migration Ring - Broad' and (optionally) adds device members to a chosen ring. These are the groups the Intune 'Firewall Policy - Migrated Rules - *' and 'Firewall Rules - Migrated * - *' profiles are assigned to. Group creation is gated behind -Apply (default is a safe preview).

How to run: Connect to Microsoft Graph first (delegated, least privilege): Connect-MgGraph -Scopes 'Group.ReadWrite.All','Device.Read.All' Preview what would be created (safe, default): .\FWMIG-Set-MigrationRingTargeting.ps1 Create the groups: .\FWMIG-Set-MigrationRingTargeting.ps1 -Apply Create groups and add devices to a ring (by Entra device object id):

.\FWMIG-Set-MigrationRingTargeting.ps1 -Apply -Ring Pilot -DeviceObjectId <id1>,<id2>

FWMIG-Validation.ps1

Purpose: Confirms a migrated device's EFFECTIVE Windows Defender Firewall state matches the captured GPO inventory (the parity baseline from FWMIG-Discovery.ps1), and that GPO is no longer contributing rules (single authority). Reports per-profile default parity, per-rule parity (present / missing / mismatched / stray), and any still-applying GPO firewall rules. This script is READ-ONLY - it makes no changes.

How to run: Run on a migrated device (in the current ring) in an elevated PowerShell session, pointing at the baseline JSON produced by FWMIG-Discovery.ps1 before cutover: .\FWMIG-Validation.ps1 -BaselinePath C:\Temp\FWMIG\FWMIG-Inventory-PC01-...json Uses the built-in NetSecurity module.